Multi-Modal Gesture Recognition Using RGB and Thermal Video: Baseline Study Across Camera Distances

Sajjan Singh[1] Aman Singh[1] Bhashkar Kashyap[1]

[1] *Dept. of Data Science and Engineering, Indian Institute of Science Education and Research (IISER) Bhopal, Bhopal, Madhya Pradesh, India*

{sajjan25, amans25, bhashkar25}@iiserb.ac.in

Abstract—Gesture recognition systems deployed in security, emergency response, and accessibility applications routinely face two violations of standard evaluation assumptions: the test-time camera–subject standoff distance differs from training, and illumination conditions change. This paper presents a systematic multi-modal baseline study addressing both issues. We collect **749 paired RGB and thermal infrared video clips** from **107 subjects** performing **seven gesture classes** (*doctor, emergency, fire, help, robbery, sit down, stand up*) at three distances: 4 ft, 6 ft, and 8 ft. Two evaluation protocols are defined: Protocol A (subject-independent split) and Protocol B (21 cross-distance splits from all combinations of training distance configurations and test distances). Five model families are evaluated: frame-level CNN baselines on RGB and background-removed RGB (BGREM), CNN+LSTM and CNN+GRU temporal models on thermal, clip-level GestFormer and VideoMAE on all three modalities, and dual-stream GestFormerFusion and ConvNeXtTinyGRU fusion architectures. A day–night generalisation experiment is also conducted. The best single-modality result is GestFormer on thermal (97.9% mean accuracy at all-distance training). Dual-stream fusion of BGREM and thermal achieves **98.9%** (GestFormerFusion) and **98.6%** (ConvNeXtTinyGRU). Cross-distance accuracy falls as low as 22% with single-source training but recovers by up to 41 pp when a second distance is added. VideoMAE fails entirely, remaining near the 14.3% chance baseline across all configurations, due to severe domain mismatch with its Kinetics-400 pretraining.

Keywords— gesture recognition, RGB–thermal fusion, cross-distance generalisation, CNN+LSTM, CNN+GRU, GestFormer, VideoMAE, subject-independent evaluation.

I. Introduction

Gesture-based communication underpins a growing set of high-stakes applications. In emergency response, a person unable to speak may still gesture. In security surveillance, specific gestures can trigger automated alerts. In accessibility technology, gesture recognition forms the basis of sign-language interpretation for the hearing-impaired [7, 8]. Despite these motivations, most deployed systems are evaluated under tightly controlled laboratory conditions where both camera distance and illumination are fixedconditions that rarely hold in deployment.

This project targets two concrete operational questions. *First*, if a gesture recognition model is trained at a fixed standoff distance, how much does accuracy degrade when the same gestures are performed at a different distance? *Second*, can a thermal infrared-based systempassive, night-capable, and insensitive to colour texture variation from clothing or skin tonetransfer better across distances and lighting conditions than an RGB-based one?

Contributions. This work provides:

- A paired RGB–thermal dataset: 749 clips, 107 subjects, 7 classes, 3 distances, with documented collection and subject-reconstruction protocols.
- Two complementary protocols: Protocol A (subject-independent) and Protocol B (21 cross-distance splits).
- A frame-level RGB benchmark across five ImageNet-pretrained backbones (ResNet-18, EfficientNet-B0, MobileNetV3-Small, ViT-B/16, ConvNeXt-Tiny) with and without background removal, across all 22 protocol splits.
- Thermal temporal models: five backbones × CNN+LSTM/GRU at layer depths 1–5, across all 21 Protocol B splits.
- Clip-level models: GestFormer and VideoMAE on RGB, BGREM, and thermal across all 21 splits.
- Dual-stream fusion: GestFormerFusion and ConvNeXtTiny-GRU with a day–night robustness study.
- Quantified zero-shot cross-modal transfer in both directions.

II. Related Work

Gesture recognition with deep learning evolved through three phases: 2D frame-level CNNs with score aggregation [1], 3D convolutional models (C3D, I3D) for joint spatiotemporal modelling [2, 3], and transformer-based clip models [4] achieving state-of-the-art on large benchmarks. Recurrent architectures combining CNN feature extraction with LSTM [5] or GRU [6] offer a parameter-efficient middle ground [7, 8].

Thermal infrared cameras detect emitted heat, making them insensitive to ambient illumination and unaffected by colour texture variation [9, 10]. The absence of discriminative texture forces thermal models to rely on body shape and motion, which improves cross-domain transfer as our results confirm.

Multi-modal fusion for RGB and thermal has been studied in pedestrian detection [11], face recognition [12], and activity recognition [13]. Bidirectional cross-modal attention [14] allows each modality to selectively attend to complementary information in the other, outperforming simple concatenation.

Background removal via instance segmentation (Mask R-CNN) or salient object detection (U2-Net [15]) can help when background texture encodes dataset-specific cues, but may discard contextual information and disturb feature normalisation in some architectures [16].

VideoMAE [17] adapts masked autoencoding to video by masking 75–90% of spatiotemporal tokens and training an encoder to reconstruct them, yielding strong representations on Kinetics-400. However, the domain gap to short isolated gesture clips in thermal creates a fundamental challenge.

III. Dataset and Protocol Design

A. Dataset Overview

The dataset consists of **749 paired clips** from **107 subjects** performing isolated dynamic gestures at three standoff distances: 4 ft, 6 ft, and 8 ft. Seven gesture classes are included: *doctor, emergency, fire, help, robbery, sit down,* and *stand up*. Each sample contains a synchronised RGB clip and a ther-

mal infrared clip from co-located cameras, preserving spatial correspondence without post-hoc registration. The dataset is approximately balanced across distances (≈238, 245, 266 clips at 4 ft, 6 ft, 8 ft respectively).

The seven classes represent emergency-relevant signalling scenarios: *doctor, emergency, help,* and *robbery* are explicit alert gestures; *fire* involves distinct pointing or waving motion; *sit down* and *stand up* are body-state transitions for crowd management scenarios. Crucially, the latter two are the most visually ambiguous pairsharing similar whole-body motion trajectories in opposite directionsmaking them a diagnostic for temporal modelling quality.

RGB clips were captured at 30 fps at 1920×1080 resolution with a standard visible-spectrum camera. Thermal clips were captured by a co-located infrared camera with the two sensors mounted adjacently to minimise parallax. The recording environment was an indoor laboratory with controlled but not completely uniform lighting.

B. Data Preparation

Pairing. RGB and thermal recordings within each (distance, gesture) group were matched by acquisition serial number, yielding a deterministic, reproducible pairing rule validated by manual inspection.

Frame extraction. FFmpeg with GPU-accelerated decoding (CPU fallback) was used. One second was removed from each end of each clip to discard setup and tear-down artefacts.

Background removal. A BGREM variant was produced per RGB frame using the `rembg` library with U2-Net [15]:

$$I_{\text{bgrem}}(i,j) = A(i,j) \cdot I(i,j) + (1 - A(i,j)) \cdot \mathbf{0}, \quad (1)$$

where $A \in [0,1]^{H \times W}$ is U2-Net's saliency mask. The result retains only the subject's body, stored as a three-channel image with a black background.

Subject identity. Subject IDs were reconstructed assuming the pair index within each distance group is consistent across all seven gesture folders. This produced a subject-id column for subject-independent splitting, validated by verifying ≈33–36 subjects per distance group.

Clip encoding. $T = 32$ frames for frame-level RGB models; $T = 10$ for LSTM/GRU thermal models (VRAM constraint: 17.2 GB RTX A4000); $T = 16$ for GestFormer and VideoMAE. Thermal frames (single-channel) are replicated to three channels for compatibility with ImageNet-pretrained encoders.

C. Protocol Design

Protocol A uses a subject-independent split (train: 525, val: 77, test: 147) pooling all distances simultaneously, measuring generalisation across people without distance shift.

Protocol B defines 21 splits from the Cartesian product of training distance configurations $C = \{\{4\},\{6\},\{8\},\{4,6\},\{4,8\},\{6,8\},\{4,6,8\}\}$ and test distances $D = \{4\text{ft}, 6\text{ft}, 8\text{ft}\}$. The three same-distance protocols serve as in-distribution upper bounds; the remaining 18 splits introduce varying covariate shift (6 single-source, 9 two-source, 3 all-source).

IV. Methodology

A. Frame-Level RGB Baselines

Five ImageNet-1K pretrained backbones are evaluated: ResNet-18 [18], EfficientNet-B0 [19], MobileNetV3-Small [20], ViT-B/16 [21], and ConvNeXt-Tiny [22]. The bottom 60% of parameters are frozen following Yosinski et al. [26].

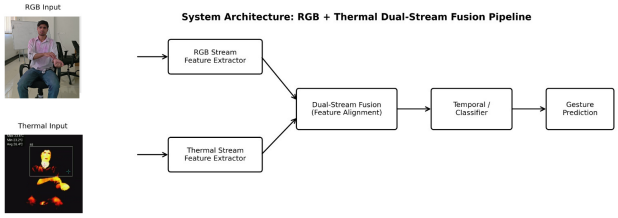

Figure 1: Dual-stream fusion pipeline. Independent RGB/BGREM and thermal EfficientNet-B0 encoders produce per-frame token sequences that are aligned by bidirectional cross-modal attention (GestFormerFusion) or concatenated for GRU-based temporal modelling (ConvNeXtTinyGRU), before a linear classification head predicts one of seven gesture classes.

For each frame $x_t \in \mathbb{R}^{3 \times H \times W}$:

$$\hat{y}_t = \text{Backbone}(x_t; \theta), \quad \hat{y}_t \in \mathbb{R}^7, \quad (2)$$

and the clip prediction is the majority vote over $T = 32$ per-frame decisions:

$$\hat{y}_{\text{clip}} = \arg\max_c \sum_{t=1}^{T} \mathbf{1}[\arg\max \hat{y}_t = c]\,. \quad (3)$$

B. CNN+LSTM and CNN+GRU for Thermal

A convolutional backbone extracts $f_t \in \mathbb{R}^{d_{\text{enc}}}$ per frame. The LSTM [5] processes the sequence $(f_1, \ldots, f_{10})$ with $d_h = 256$:

$$i_t = \sigma(W_i[h_{t-1}; f_t] + b_i), \qquad f_t^g = \sigma(W_f[h_{t-1}; f_t] + b_f), \quad (4)$$

$$g_t = \tanh(W_g[h_{t-1}; f_t] + b_g), \quad o_t = \sigma(W_o[h_{t-1}; f_t] + b_o), \quad (5)$$

$$c_t = f_t^g \odot c_{t-1} + i_t \odot g_t, \qquad h_t = o_t \odot \tanh(c_t). \quad (6)$$

The GRU [6] merges input and forget gates into a single update gate:

$$r_t = \sigma(W_r[h_{t-1}; f_t] + b_r), \quad u_t = \sigma(W_u[h_{t-1}; f_t] + b_u), \quad (7)$$

$$\tilde{h}_t = \tanh(W_h[(r_t \odot h_{t-1}); f_t] + b_h), \quad h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \tilde{h}_t. \quad (8)$$

The final hidden state $h_T \in \mathbb{R}^{256}$ feeds a linear head $\hat{y} = W_c h_T + b_c$. Layer depth $L \in \{1,2,3,4,5\}$ is treated as a hyperparameter.

C. GestFormer

GestFormer encodes each of $T = 16$ frames with EfficientNet-B0 and pools spatially: $f_t = \text{GAP}(\text{EfficientNet-B0}(x_t)) \in \mathbb{R}^{1280}$. Each f_t is projected to $d = 256$, layer-normalised, and augmented with a learnable `[CLS]` token and sinusoidal positional encoding [23]:

$$\text{PE}(\text{pos}, 2k) = \sin\left(\frac{\text{pos}}{10000^{2k/d}}\right), \ \text{PE}(\text{pos}, 2k{+}1) = \cos\left(\frac{\text{pos}}{10000^{2k/d}}\right). \quad (9)$$

The $T+1$ token sequence is processed by a 4-layer pre-norm transformer [24] ($H = 8$ heads, FFN dim 512); the `[CLS]` output is classified by a linear head. Label smoothing $\epsilon = 0.1$ is applied: $\tilde{y}_c = (1-\epsilon)\mathbf{1}[c = y] + \epsilon/K$. GestFormer is evaluated independently on RGB, BGREM, and thermal.

D. Dual-Stream Fusion

GestFormerFusion (GF) extends GestFormer to a 6-channel input. Two EfficientNet-B0 encoders produce token sequences $Z^{\text{rgb}}, Z^{\text{thm}} \in \mathbb{R}^{T \times 256}$, aligned by bidirectional cross-modal attention [14]:

$$\hat{Z}^{\text{rgb}} = \text{MHA}(Z^{\text{rgb}}, Z^{\text{thm}}, Z^{\text{thm}}), \quad (10)$$

$$\hat{Z}^{\text{thm}} = \text{MHA}(Z^{\text{thm}}, Z^{\text{rgb}}, Z^{\text{rgb}}). \quad (11)$$

Table 1: Training hyperparameters for all model families.

Parameter	Frame-RGB	LSTM/GRU	GestFormer	VideoMAE
Optimiser	Adam	Adam	AdamW	AdamW
LR	10^{-4}	10^{-4}	5×10^{-4}	10^{-4}
Weight decay	—	—	0.01	0.05
Batch size	10–24	8	32	16
Max epochs	10–30	100	60	50
Early stop pat.	7	7	15	12
Frames T	32	10	16	16
Backbone freeze	60%	60%	none	full init
Unfreeze epoch	—	—	—	10
LR schedule	Plateau	none	Cosine	Cosine
Label smoothing	—	—	0.1	—

Each output is refined by a residual FFN (hidden $2d = 512$, GELU), then merged:

$$Z^{\text{fused}} = \text{LN}(\tilde{Z}^{\text{rgb}} + \tilde{Z}^{\text{thm}}). \tag{12}$$

The fused sequence (with prepended `[CLS]` and sinusoidal PE) is processed by a 4-layer pre-norm transformer and a linear head.

ConvNeXtTinyGRU (CX) uses two ConvNeXt-Tiny [22] encoders (60% frozen) producing 768-dim features per frame. Early fusion is performed by concatenation [25]:

$$h_t = [g_t^{\text{rgb}};\, g_t^{\text{thm}}] \in \mathbb{R}^{1536}, \tag{13}$$

processed by a 1-layer GRU ($d_h = 256$) whose last hidden state feeds a linear head. In contrast to GestFormerFusion's explicit cross-modal alignment, CX learns to weight the two streams implicitly through GRU gating.

E. Training Configuration

All models use cross-entropy loss. Optimisation details are in Table 1. RGB frame-level and cross-modal experiments ran on an NVIDIA H100 NVL (99.9 GB VRAM); thermal and fusion models on an NVIDIA RTX A4000 (17.2 GB VRAM). Data augmentation for RGB includes random horizontal flip, random crop with padding, and colour jitter. Thermal augmentation omits colour jitter.

V. Experiments and Results

A. RGB Frame-Level Results

Table 2 reports Protocol A results. EfficientNet-B0 achieves the highest raw-RGB accuracy (88.4%, F1 88.5%); ConvNeXt-Tiny leads on BGREM (88.4%, F1 88.3%). The two architectures are within 0.1 pp on Protocol A, but diverge sharply under cross-distance shift.

Table 2: Protocol A frame-level accuracy (%) and macro-F1 (%). **Bold**: best per column.

	RGB		BGREM	
Backbone	Acc	F1	Acc	F1
ResNet-18	81.0	80.8	83.0	82.8
EfficientNet-B0	**88.4**	**88.5**	81.6	82.0
MobileNetV3-Small	75.5	75.8	51.0	48.2
ViT-B/16	57.8	52.9	83.7	83.1
ConvNeXt-Tiny	84.4	84.6	**88.4**	**88.3**

ViT-B/16 shows the most dramatic sensitivity to background removal: +25.9 pp (57.8%→83.7%). Lacking local inductive bias, ViT over-relies on global background statistics. Removing the background forces attention to body shapethe true discriminative feature for gesture. MobileNetV3-Small shows the opposite: −24.5 pp (75.5%→51.0%), since its depthwise separable convolutions are sensitive to low-level texture statistics disrupted by the large black regions introduced by background removal.

Per-class F1 for EfficientNet-B0 on Protocol A: *doctor* 97.7%, *emergency* 90.5%, *fire* 90.9%, *help* 97.6%, *robbery* 89.5%, *sit down* 74.4%, *stand up* 79.1%. The two lowest classes*sit down* and *stand up*share similar whole-body posture transitions that frame-level majority voting cannot distinguish without temporal context.

Table 3 reports cross-distance results for ConvNeXt-Tiny and EfficientNet-B0. ConvNeXt-Tiny achieves a mean of 87.2% across 22 protocols versus 78.4% for EfficientNet-B0. At 8 ft→6 ft, ConvNeXt-Tiny scores 73.1% vs. EfficientNet-B0's 38.8% (+34.3 pp); at 8 ft→4 ft, 54.2% vs. 30.7%. Large-kernel (7×7) depthwise convolutions in ConvNeXt-Tiny produce more distance-invariant spatial features than EfficientNet-B0's channel attention.

Table 3: Cross-distance accuracy (%) for ConvNeXt-Tiny and EfficientNet-B0 across all 22 protocols. **Bold**: highest per row.

		RGB		BGREM	
Group	Protocol	CX	EB0	CX	EB0
Subj-indep	Protocol A	84.4	**88.4**	**88.4**	81.6
Same	4ft→4ft	**98.7**	95.0	97.5	96.6
	6ft→6ft	97.6	**98.8**	98.0	98.0
	8ft→8ft	**97.0**	88.0	95.5	95.9
Single source	4ft→6ft	**85.3**	55.5	84.5	75.9
	4ft→8ft	60.2	42.9	**66.9**	51.5
	6ft→4ft	**81.5**	63.9	80.3	70.2
	6ft→8ft	64.3	39.8	**66.2**	56.0
	8ft→4ft	**54.2**	30.7	55.0	36.1
	8ft→6ft	**73.1**	38.8	68.2	51.8
Two source	4ft+6ft→4ft	95.4	**97.1**	95.8	96.2
	4ft+6ft→6ft	97.1	**98.4**	95.9	98.0
	4ft+6ft→8ft	**78.6**	62.8	75.9	62.4
	4ft+8ft→6ft	**88.2**	80.8	87.3	82.0
	6ft+8ft→4ft	**87.0**	74.4	81.1	73.1
	6ft+8ft→6ft	**99.6**	98.8	99.2	98.8
All source	All→4ft	**95.0**	93.3	95.0	**99.2**
	All→6ft	**98.0**	96.3	91.4	93.5
	All→8ft	**94.0**	93.6	88.7	89.1
Mean (all 22)		**87.2**	78.4	86.4	80.7

B. Thermal Temporal Models: CNN+LSTM and CNN+GRU

Table 4 reports results at all-distance training. ConvNeXt-Tiny+GRU achieves the best mean (80.2%). CNN+GRU consistently matches or exceeds CNN+LSTM for the same backboneconsistent with theory: on short sequences ($T = 10$) the GRU's reduced parameter count is less prone to overfitting, and merged gating does not sacrifice capacity.

Table 4: CNN+LSTM and CNN+GRU thermal accuracy (%) at all-distance training. Best layer depth selected. M = mean.

	CNN+LSTM				CNN+GRU			
Backbone	4ft	6ft	8ft	M	4ft	6ft	8ft	M
ResNet-50	46.9	48.2	64.3	53.1	55.1	57.1	60.7	57.6
EfficientNet-B0	63.3	71.4	73.2	69.3	69.4	69.6	73.2	70.7
MobileNetV3	67.3	62.5	69.6	66.5	63.3	62.5	73.2	66.3
ViT-B/16	81.6	73.2	75.0	76.6	71.4	73.2	75.0	73.2
ConvNeXt-Tiny	69.4	75.0	76.8	73.7	**81.6**	75.0	**83.9**	**80.2**

The GRU depth ablation (Table 5) reveals that a single GRU layer is definitively optimal. Depth 2 collapses to 24–27%barely above chance. This is a discrete failure: the 2-layer GRU adds a recurrent pathway with effective gradient length $T \times L = 10 \times 2 = 20$, which the 749-sample dataset cannot support reliably. Layers 3–5 partially recover but do not surpass depth 1. The LSTM depth ablation shows a similar pattern; LSTM depth 4 edges out depth 1 by 0.7 pp for ConvNeXt-Tiny (74.4% vs. 73.7%), as the separate cell state buffers gradient flow.

C. Cross-Modal Transfer

RGB-trained models tested on thermal without retraining cluster near 20% (chance: 14.3%), with ConvNeXt-Tiny the sole exception at 41.2% mean (Table 6). ConvNeXt-Tiny's shape-sensitive large-kernel convolutions learn edge and silhouette

Table 5: GRU layer depth ablation: ConvNeXt-Tiny, all-distance training.

Layers	4ft (%)	6ft (%)	8ft (%)	Mean (%)
1	**81.6**	73.2	**83.9**	**79.6**
2	24.5	25.0	26.8	25.4
3	77.6	**75.0**	71.4	74.7
4	73.5	69.6	78.6	73.9
5	75.5	71.4	69.6	72.2

features present in both modalities; EfficientNet-B0's SE attention is calibrated to RGB channel statistics and provides no useful thermal signal.

In the reverse direction, thermal-trained models reach 63–80% on RGB without adaptation: ConvNeXt-Tiny+LSTM achieves 80.4% at 6 ft RGB. This directional asymmetry is a key finding: thermal training forces body-shape-centred features that generalise across modality; RGB training encodes colour and texture statistics absent in thermal.

Table 6: Zero-shot cross-modal transfer accuracy (%).

Direction	Model	4ft	6ft	8ft
RGB→Thm	ConvNeXt-Tiny	47.9	46.9	27.8
	EfficientNet-B0	24.5	18.0	19.6
Thm→RGB	ConvNeXt+LSTM	**73.5**	**80.4**	73.2
	ConvNeXt+GRU	63.3	78.6	71.4

D. GestFormer and VideoMAE

GestFormer-Thermal achieves the highest mean across 21 Protocol B splits (83.2%), outperforming GestFormer-BGREM (80.8%) and GestFormer-RGB (79.6%). The thermal advantage is largest on the hardest protocols: at 8 ft→6 ft, thermal scores 58.4% vs. RGB's 22.0% (+36.4 pp). Even on same-distance protocols, GestFormer-Thermal surpasses RGB (97.1% vs. 89.5% at 4 ft→4 ft), indicating that for this dataset thermal provides cleaner, more consistent gesture features regardless of distance shift.

VideoMAE produces near-chance accuracy across all 21 splits and all three modalities (mean: RGB 17.2%, BGREM 23.0%, thermal 18.9%). Three compounding causes explain this failure: *(i)* temporal statistics mismatchKinetics-400 contains 10-second natural videos while our clips are <2 s after trimming; *(ii)* spatial mismatchthe masked autoencoding objective trains the encoder to reconstruct background texture, which is trivially static in our single indoor environment and provides no useful gradient; *(iii)* parameter-to-sample ratioVideoMAE-Base has 86 M parameters fine-tuned from at most 651 samples (≈75K parameters/sample).

E. Dual-Stream Fusion Results

Table 7 reports dual-stream fusion results. GF-BGREM+T wins on 15/21 protocols (mean 86.8% vs. 84.4% for CX-BGREM+T, +2.4 pp). The cross-modal attention mechanism explicitly aligns RGB and thermal tokens before temporal modelling. Background removal is beneficial in fusion: GF-BGREM+T outperforms GF-RGB+T on 16/21 protocols (mean +3.2 pp), with the largest gain at 8 ft→4 ft (+10.9 pp: 26.1%→37.0%). Removing background discards distance-specific scene texture, making the RGB and thermal feature spaces more comparable for cross-modal attention.

CX-BGREM+T leads on the two hardest far-to-near protocols: 8 ft→6 ft (68.6% vs. 46.1%, +22.5 pp) and 8 ft→4 ft (42.0% vs. 37.0%, +5.0 pp). Here the two modalities have highly asymmetric qualitythe RGB stream is severely out-of-distributionand the GRU's implicit gating learns to down-weight it and rely on thermal, something the cross-modal attention mechanism does less adaptively on a small training set.

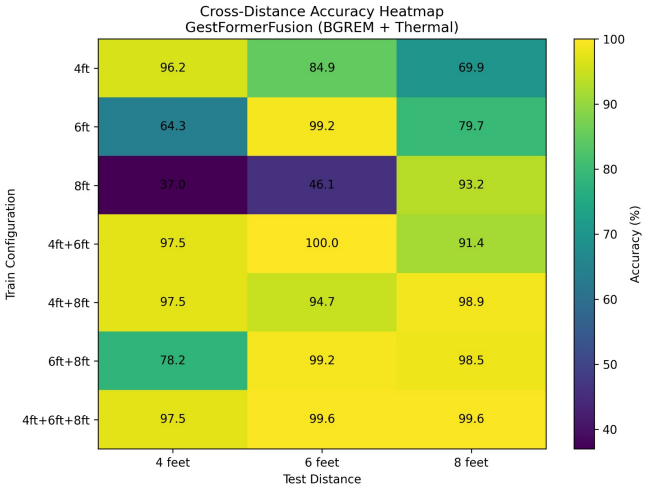

Figure 2: Cross-distance accuracy heatmap for GestFormerFusion (BGREM+Thermal) across all 21 Protocol B splits. Rows: training distance configurations; columns: test distances. Darker green indicates higher accuracy. 8 ft-only training produces the worst transfer; all-distance training recovers near-ceiling performance.

Adding one training distance recovers 30–41 pp: GF-BGREM+T at 6 ft+8 ft→4 ft reaches 78.2% versus 37.0% with 8 ft-only (+41.2 pp). 8 ft-only training is the worst source across all model families: 8 ft→4 ft ranges from 26–44%, while 4 ft→8 ft ranges from 60–72%. This asymmetry reflects the direction of scale change: training at small pixel count (8 ft) and testing at large pixel count (4 ft) requires handling unseen feature scales, which is harder than the reverse.

Table 7: Dual-stream fusion accuracy (%) across selected Protocol B splits. GF = GestFormerFusion; CX = ConvNeXtTinyGRU. Unimodal GestFormer-Thermal shown for reference. **Bold**: best fusion result per row.

Grp	Protocol	GF-Thm (unimod)	GF RGB+T	GF BGR+T	CX RGB+T	CX BGR+T
Same	4ft→4ft	97.1	95.4	**96.2**	76.1	83.2
	6ft→6ft	94.3	99.2	**99.2**	94.7	95.9
	8ft→8ft	90.6	85.7	93.2	95.9	**96.6**
Sngl src	4ft→6ft	77.6	80.8	**84.9**	72.7	80.0
	4ft→8ft	62.4	63.5	69.9	69.9	**71.8**
	6ft→4ft	53.4	60.1	**64.3**	59.2	55.0
	6ft→8ft	70.7	72.9	**79.7**	73.7	76.3
	8ft→4ft	41.6	26.1	37.0	**44.1**	42.0
	8ft→6ft	58.4	36.3	46.1	66.9	**68.6**
Two src	4ft+6ft→4ft	97.5	97.5	**97.5**	93.7	90.8
	4ft+6ft→6ft	98.0	**100.**	**100.**	99.2	98.0
	4ft+6ft→8ft	80.8	87.2	**91.4**	83.5	80.1
	4ft+8ft→6ft	87.8	93.1	**94.7**	91.8	89.4
	6ft+8ft→4ft	66.0	72.7	**78.2**	70.2	65.6
	6ft+8ft→8ft	90.6	99.3	98.5	97.0	**98.1**
All src	All→4ft	97.1	96.2	**97.5**	96.6	96.2
	All→6ft	98.0	99.2	**99.6**	98.8	**99.6**
	All→8ft	98.5	98.5	99.6	**99.6**	**100.**
Mean (21 splits)		83.2	83.6	**86.8**	84.6	84.4

F. Day–Night Generalisation

GestFormerFusion drops 15.2 pp from RGB day→RGB night (83.3%→68.1%), while ConvNeXtTinyGRU drops only 1.5 pp (82.0%→80.5%). EfficientNet-B0's SE channel attention builds statistics on global average-pooled features that change substantially with ambient illumination; ConvNeXt-Tiny's 7×7 depthwise convolutions respond to local edge patterns less sensitive to global brightness shifts.

Training on nighttime RGB and testing on daytime RGB degrades both models further (GestFormerFusion: 56.6%; ConvNeXtTinyGRU: 66.5%), since nighttime training encodes reduced-contrast, low-saturation featuresa strict subset of daytime appearance variation.

Thermal imaging is nearly illumination-invariant: Con-

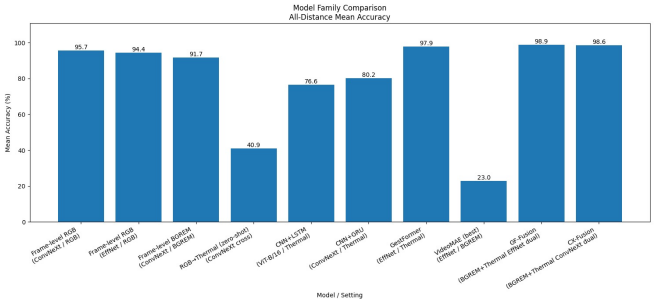

Figure 3: Mean test accuracy (%) per model family at all-distance training, evaluated across 4 ft, 6 ft, and 8 ft test sets. GestFormerFusion with BGREM+Thermal achieves the highest mean (98.9%), followed by ConvNeXtTinyGRU (98.6%). VideoMAE remains near the 14.3% chance baseline.

vNeXtTinyGRU degrades by only 2.2 pp thermal day→thermal night (72.0%→69.8%). GestFormerFusion shows a larger drop of 6.8 pp (68.5%→61.7%) because EfficientNet-B0's spatial encoder is more sensitive to input statistics than ConvNeXt-Tiny's locally focused depthwise convolutions.

G. All-Distance Training Summary

Figure 3 and Table 8 compare all model families at all-distance training. Frame-level RGB models achieve 91–96% mean, substantially above thermal temporal models (76–80%), because RGB provides richer per-frame spatial detail. GestFormer-Thermal (97.9%) surpasses frame-level RGB (95.7%) by combining stable thermal features with transformer temporal modelling. Dual-stream fusion adds ≈1 pp over the best unimodal baseline, consistently across most protocols.

Table 8: All-distance training summary: mean test accuracy (%) across 4 ft, 6 ft, 8 ft. Best configuration per family.

Family	Input	4ft	6ft	8ft	Mean
Frame-level RGB	ConvNeXt / RGB	95.0	98.0	94.0	95.7
Frame-level RGB	EffNet / RGB	93.3	96.3	93.6	94.4
Frame-level BGREM	ConvNeXt / BGREM	95.0	91.4	88.7	91.7
RGB→Thm (zero-shot)	ConvNeXt	47.9	46.9	27.8	40.9
CNN+LSTM	ViT-B/16 / Thermal	81.6	73.2	75.0	76.6
CNN+GRU	ConvNeXt / Thermal	81.6	75.0	83.9	80.2
GestFormer	EffNet / Thermal	97.1	98.0	98.5	97.9
VideoMAE	EffNet / BGREM	24.0	25.7	19.2	23.0
GF-Fusion	BGREM+Thm	97.5	99.6	99.6	**98.9**
CX-Fusion	BGREM+Thm	96.2	99.6	100.	**98.6**

VI. Conclusion

This project systematically characterises multi-modal gesture recognition under the two most practically important forms of distribution shift: standoff distance and illumination change. We evaluated five model families across 22 protocols on a 749-clip paired RGB–thermal dataset of seven gesture classes from 107 subjects.

Key findings:

- **Distance shift is severe but recoverable.** Single-source cross-distance accuracy falls to 22–27%; adding one training distance recovers 30–41 pp. All-distance dual-stream fusion reaches 96–100%.
- **Thermal is the more generalisable modality.** GestFormer-Thermal outperforms GestFormer-RGB across all 21 Protocol B splits on average; the advantage is largest on the hardest far-to-near protocols.
- **Modality transfer is asymmetric.** Thermal training transfers to RGB (63–80% zero-shot); RGB training does not transfer to thermal (≈20%). Thermal features are body-shape centred and modality-agnostic; RGB features encode colour/texture statistics absent in thermal.
- **Background removal is architecture-dependent.** Global-attention models (ViT-B/16) gain +25.9 pp; texture-dependent models (MobileNetV3-Small) lose 24.5 pp. In fusion, BGREM+Thermal consistently outperforms RGB+Thermal.
- **GRU depth 1 is optimal.** Depth 2 collapses to near-chance (24–27%); the 749-sample dataset cannot support longer recurrent gradients.
- **VideoMAE is inapplicable.** 86 M parameters cannot be adapted from Kinetics-400 to 749 short clips without severe domain mismatch in temporal statistics, spatial content, and parameter-to-sample ratio.
- **ConvNeXtTinyGRU is preferred for hard far-to-near transfer.** GRU implicit gating learns to downweight the degraded out-of-distribution RGB stream, outperforming the explicit cross-modal attention of GestFormerFusion on the 8 ft→4 ft and 8 ft→6 ft protocols.

Limitations. The dataset (749 clips) is small relative to standard action recognition benchmarks. Subject-identity reconstruction relies on serial-number ordering and may have residual errors. The day–night experiment uses a subset of clips; exact illumination conditions are not documented in published detail.

References

[1] K. Simonyan and A. Zisserman, "Two-stream convolutional networks for action recognition in videos," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 27, 2014.

[2] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, pp. 4489–4497, 2015.

[3] J. Carreira and A. Zisserman, "Quo vadis, action recognition? A new model and the Kinetics dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 6299–6308, 2017.

[4] G. Bertasius, H. Wang, and L. Torresani, "Is space-time attention all you need for video understanding?" in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 139, pp. 813–824, 2021.

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[6] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, pp. 1724–1734, 2014.

[7] L. Pigou, A. van den Oord, S. Dieleman, M. Van Herreweghe, and J. Dambre, "Beyond temporal pooling: Recurrence and temporal convolutions for gesture recognition in video," *Int. J. Comput. Vis.*, vol. 126, no. 2, pp. 430–439, 2018.

[8] Q. Miao et al., "Multimodal gesture recognition based on the ResC3D network," in *Proc. IEEE Int. Conf. Comput. Vis. Workshop (ICCVW)*, pp. 3047–3055, 2017.

[9] R. Gade and T. B. Moeslund, "Thermal cameras and applications: A survey," *Mach. Vis. Appl.*, vol. 25, no. 1, pp. 245–262, 2014.

[10] O. Kopuklu, A. Gunduz, N. Kose, and G. Rigoll, "Real-time hand gesture detection and classification using convolutional neural networks," in *Proc. IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, pp. 1–8, 2019.

[11] W. Liu et al., "Multispectral pedestrian detection: Benchmark dataset and baseline," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 1037–1045, 2016.

[12] Z. Shao, B. Ma, B. Liu, and H. Zhang, "3D face recognition using thermal and visual imagery," in *Proc. IEEE Int. Symp. Biometrics and Security Technol. (ISBA)*, pp. 1–6, 2014.

[13] E. Ohn-Bar and M. M. Trivedi, "Hand gesture recognition in real time for automotive interfaces," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 6, pp. 2368–2377, 2014.

[14] H. Zhang et al., "CM-Diff: Cross-modal diffusion with fused cross-modal features for video generation," *arXiv preprint arXiv:2503.09514*, 2025.

[15] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O. R. Zaiane, and M. Jagersand, "U^2-Net: Going deeper with nested U-structure for salient object detection," *Pattern Recognition*, vol. 106, p. 107404, 2020.

[16] Y. Sheikh and M. Shah, "Bayesian modeling of dynamic scenes for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 11, pp. 1778–1792, 2005.

[17] Z. Tong, Y. Song, J. Wang, and L. Wang, "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 770–778, 2016.

[19] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, pp. 6105–6114, 2019.

[20] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, pp. 1314–1324, 2019.

[21] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[22] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 11976–11986, 2022.

[23] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.

[24] R. Xiong et al., "On layer normalization in the transformer architecture," in *Proc. Int. Conf. Mach. Learn. (ICML)*, pp. 10524–10533, 2020.

[25] C. G. M. Snoek, M. Worring, and A. W. M. Smeulders, "Early versus late fusion in semantic video analysis," in *Proc. ACM Int. Conf. Multimedia*, pp. 399–402, 2005.

[26] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?" in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 27, 2014.

[27] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.

[28] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[29] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, "A theory of learning from different domains," *Mach. Learn.*, vol. 79, no. 1, pp. 151–175, 2010.